

Project Charter

Hardwood Legacy App

Team Member:

Luke Smith

Problem Statement:

The college basketball world has been looking for a game that collectively encapsulates the college basketball landscape, with ever-changing aspects like NIL and the transfer portal. Right now, there is no way to experience what it is like to be a coach, go through the process of recruiting and building a great college program, without actually being a coach. Hardwood Legacy looks to change that by providing a realistic college basketball experience that will make casual and die-hard fans alike happy. We look to allow decades of simulation, finding hidden gems, and adding all of the niche details that make college basketball so much fun for fans.

Project Objectives:

1. Create a realistic college basketball simulation experience
2. Present real-time simulation statistics in a format that makes the experience feel authentic and immersive.
3. Have an easily accessible platform where users can understand information and know where to go for it.
4. Simulate recruiting techniques such as visits, scouting, NIL, and other tactics
5. Develop a successful prototype for a Q4 launch as a solo developer

Stakeholders:

Users: College basketball fans and creators

Developer: Luke Smith

Tester: Luke Smith

Project Owner: Luke Smith

Deliverables:

- Game planning page to coach your team and establish strategies against the opposing team
- Realistic statistics to analyze performance and evaluate potential prospects
- Scouting and recruiting that mirrors processes used by college coaches around the country
- Realistic characteristics of a university, basing recruiting and performance on funding, prestige, fan support, and other aspects

Tech Stack:

- React Native front-end for iOS and Android
- FastAPI + Celery Backend for long-running simulation and API handling
- Python simulation engine for easier statistical methods
- Supabase database
- Caching through Redis
- WebSockets for live play-by-play analysis

Constraints/Risks:

- Solo App Developer => Give ample time for certain features and write easily maintainable code
- App Store review delays => submit 4-6 weeks before targeted launch
- Complex Logic for simulation => Python engine separate and independently testable
- Capability of simulating many years into the future => Use Celery to isolate and test the capability independently

Success Criteria:

- Users can simulate an accurate and realistic season
- Core features are functional and tested
- App successfully submitted to App Store and Google Play in Q4